Amritanshu Kumar

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Professional Summary

Game Engineer who ships end-to-end — client, server, API, and deployment pipeline. Currently leading a production multiplayer game across the complete stack, from Unity gameplay to TypeScript backend to Google Play. Built a game engine in C++ from first principles. Shipped titles with **5M+ downloads**.

Professional Experience

RootHoot Pvt Ltd **Remote, Hyderabad**
Team Lead *Oct 2024 – Present*

- Own the complete technical stack for Pongz, a production multiplayer Mahjong game — **14K+ players**, **2,800+ DAU**, and **62K+ matches** in early access, from Unity 6 gameplay to backend services to Play Store release pipeline.
- Solely built the real-time game server in TypeScript/Socket.IO, handling matchmaking, reconnection, AFK detection with automatic bot stand-ins, and game-state resolution to maintain fair play across unstable connections.
- Designed the ranked matchmaking and 34-tier skill rating system — **7,400+ ranked players** actively competing, balancing competitive integrity with player retention.
- Built and deployed the REST API (Express, MongoDB) with OAuth, 2FA, and rate limiting — containerized with Docker on Google Cloud Run.
- Developed 8 reusable Unity systems (UI, Event Bus, State Machine, Logging, Save, Pause, Popup, Loading) adopted across team projects, cutting setup time for new development.

DeftSoft **Mohali, Punjab**
Game Developer *Feb 2023 – Oct 2024*

- Shipped an endless runner to **5M+ downloads** and **4.6 stars** on Google Play — improved engagement **30%** by iterating on live player feedback.
- Introduced test-driven development across the team, improving user ratings by **1.5 stars** through higher release quality.
- Cut level design time by **40%** by building procedural generation tools so designers could focus on creative work instead of manual placement.
- Optimized the rendering pipeline, achieving **25%** higher frame rates on the low-end devices most players use.
- Mentored 3 junior developers — code reviews, pair programming, and establishing team coding standards.

Selected Projects

Nimirta Engine: Custom 2D game engine in C++/SFML, built from first principles — physics simulation (collision detection, impulse resolution), ECS architecture, material system, and AI opponents across 3 difficulty levels.

Recurve 28: Hardware-integrated archery game — camera, Arduino, and projector linked through OpenCV for real-time arrow impact detection. Multiple gameplay modes designed around the detection system.

Technical Skills

- **Languages:** C#, TypeScript, C++, Go, Python
- **Game Engines:** Unity, Unreal Engine, Custom (C++)
- **Backend:** Node.js, Express, MongoDB, Redis, Socket.IO, Docker, Google Cloud Run
- **Specializations:** Game Architecture, Real-Time Multiplayer, CI/CD, TDD, Clean Architecture, ECS

Education & Certifications

Lovely Professional University **Punjab, India**
Bachelor of Computer Application *Sep 2020 – Jun 2023*
Certifications: Unity Certified Programmer | Unreal Engine Game Design & Development Specialization